

DR. MOHD SAKIB

Assistant Professor, School of Computer Science
UPES, Dehradun, India

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RESEARCH INTERESTS

My research focuses on ensemble and hybrid deep learning approaches for complex time series forecasting, with applications in energy demand and smart grid systems. I develop architectures that combine signal decomposition, optimization techniques, and neural networks to improve prediction accuracy and robustness. A key challenge in my work is modeling long-range dependencies and handling evolving data distributions. My long-term goal is to design scalable and interpretable next-generation time series models aligned with emerging trends such as transformer-based architectures and foundation models for temporal data.

CURRENT POSITION

Assistant Professor

June 2025 – Present

School of Computer Science
UPES, Dehradun, India – 248007

EXPERIENCE

Assistant Professor

August 2024 – June 2025

Department of Computer Applications
Madanapalle Institute of Technology and Science (MITS Deemed to be University)
Madanapalle, Andhra Pradesh, India – 517325

Research Assistant

June 2016 – July 2017

Centre of Advanced Research in Electrified Transportation (CARET)
AMU Research Centre, Aligarh, Uttar Pradesh, India – 202002

Project: Machine learning-based day-ahead market energy usage bidding for smart microgrids

EDUCATION

Ph.D. in Computer Science

April 2021 – December 2024

Aligarh Muslim University, Aligarh, India

Supervisor: Prof. Suhel Mustajab

Thesis: An Efficient Ensemble Deep Learning Model for Estimation and Forecasting of Time Series Data

Master of Computer Applications

July 2015 – December 2018

Aligarh Muslim University, Aligarh, India

Dissertation: Mess Management System: A Case Study of SS Hall North

CGPA: 7.93/10

Bachelor of Computer Applications

June 2012 – July 2015

Aligarh Muslim University, Aligarh, India

Dissertation: Design and Development of an Online Complaint Management System

Percentage: 75.6%

Journal Articles

- **Sakib, M.**, Mustajab, S. *CEEMDAN-TCN-AutoLSTM: A Triple-Force Ensemble Using Cross-Stitch Networks for Short-Term Load Forecasting*. Arab J Sci Eng (2025). [SCI/SCIE, SCOPUS, IF:2.9, Q1]. [doi:10.1007/s13369-025-10398-1](https://doi.org/10.1007/s13369-025-10398-1)
- **Sakib, M.**, Mustajab, S., & Alam, M. *Ensemble deep learning techniques for time series analysis: a comprehensive review, applications, open issues, challenges, and future directions*. Cluster Computing, 28(1) (2025). [SCI/SCIE, SCOPUS, IF:3.6, Q1]. [doi:10.1007/s10586-024-04684-0](https://doi.org/10.1007/s10586-024-04684-0)
- **Sakib, M.**, Siddiqui, T., Mustajab, S., Alotaibi, R. M., Alshareef, N. M., & Khan, M. Z. *An ensemble deep learning framework for energy demand forecasting using genetic algorithm-based feature selection*. PLoS One, 20(1) (2025). [SCIE, SCOPUS, IF:3.7, Q1]. [doi:10.1371/journal.pone.0310465](https://doi.org/10.1371/journal.pone.0310465)
- **Sakib, M.**, Ahmad, S., Anwar, K., & Saqib, M. *Optimizing support vector regression using grey wolf optimizer for enhancing energy efficiency and building prototype architecture*. Cluster Computing, 28(1), 60 (2025). [SCI/SCIE, SCOPUS, IF:3.6, Q1]. [doi:10.1007/s10586-024-04802-y](https://doi.org/10.1007/s10586-024-04802-y)
- **Sakib, M.**, & Siddiqui, T. *Multi-network-based ensemble deep learning model to forecast Ross River virus outbreak in Australia*. International Journal of Pattern Recognition and Artificial Intelligence, 37(10) (2023). [SCI/SCIE, SCOPUS, IF:1.5, Q3]. [doi:10.1142/S0218001423520158](https://doi.org/10.1142/S0218001423520158)
- **Sakib, M.**, & Mustajab, S. *Enhanced multi-variate time series prediction through statistical-deep learning integration: the VAR-stacked LSTM model*. SN Computer Science, 5(5), 573 (2024). [SCOPUS]. [doi:10.1007/s42979-024-02950-x](https://doi.org/10.1007/s42979-024-02950-x)
- **Sakib, M.**, Usama, S., & Abhay, P. S. *Bitcoin Price Prediction using Machine Learning*. Journal of Computer Science and Software Development, 1:1–10 (2022). [doi:10.17303/jcssd.2022.1.104](https://doi.org/10.17303/jcssd.2022.1.104)
- **Sakib, M.** *E³-TSF: A Dynamic Gated Multi-Expert Transformer Framework for Robust Multi-Domain Time Series Forecasting*. Submitted to *IEEE Access*; presently under full consideration for publication.
- **Sakib, M.**, Singh, K., & Agarawal, P. *Artificial Intelligence Techniques for Cybersecurity: A Comprehensive Systematic Review of Methods, Applications, Challenges, and Research Directions*. Submitted to *Engineering Applications of Artificial Intelligence* as a survey paper.

Conference Proceedings

- **Sakib, M.**, Mustajab, S., & Siddiqui, T. *Deep learning-based heartbeat classification of 12-lead ECG time series signal*. In *2023 4th International Conference on Data Analytics for Business and Industry (ICDABI-2023)*, pp. 273–278. IEEE. [SCOPUS Indexed]. [doi:10.1109/ICDABI60145.2023.10629531](https://doi.org/10.1109/ICDABI60145.2023.10629531)
- **Sakib, M.**, & Siddiqui, T. *Anomaly detection of ECG time series signal using autoencoders neural network*. In *2023 7th International Conference On Computing, Communication, Control And Automation (ICCUBEA)*, pp. 1–7. IEEE. [SCOPUS Indexed]. [doi:10.1109/ICCUBEA58933.2023.10392094](https://doi.org/10.1109/ICCUBEA58933.2023.10392094)
- **Sakib, M.**, Mustajab, S., Anwar, K., & Alam, M. *BiLSTM-based approach for Energy Demand Forecasting and Synchronization among Smart Grids*. In *2024 IEEE Third International Conference on Power Electronics, Intelligent Control and Energy Systems (ICPEICES)*, pp. 969–974. IEEE. [SCOPUS Indexed]. [doi:10.1109/ICPEICES62430.2024.10719085](https://doi.org/10.1109/ICPEICES62430.2024.10719085)
- Khan, M., **Sakib, M.**, Siddiqui, T., & Awasthi, S. *Long-Horizon Forecasting for Sensor Systems: An Experimental Study Using Linear and Transformer-Based Models*. In *2026 Second International Conference on Emerging Computational Intelligence (ICECI)*, pp. 1–6. IEEE. [SCOPUS Indexed]. [doi:10.1109/ICECI69159.2026.11519399](https://doi.org/10.1109/ICECI69159.2026.11519399)
- **Sakib, M.**, Khan, M., Bhatt, A. K., & Vyas, S. *X-TimeAD: An Explainable Transformer-Based Framework for Multivariate Time-Series Anomaly Detection*. Submitted to an IEEE conference.

- **Sakib, M.** et al. *LLM-Guided Dynamic Expert Fusion for Multivariate Time Series Forecasting*. Submitted to an IEEE conference.
- **Sakib, M.** et al. *Security-Aware Multi-View Graph Contrastive Learning for Network Intrusion Detection*. Submitted to an IEEE conference.

Book Chapters

- Saqib, M., Sahu, S. A., **Sakib, M.**, & Al-Ammar, E. A. *Machine learning-based day-ahead market energy usage bidding for smart microgrids*. In *Electric Vehicle Integration in a Smart Microgrid Environment*, pp. 249–266. CRC Press (Taylor & Francis) (2021). [SCOPUS Indexed]. [doi:10.1201/9780367423926-10](https://doi.org/10.1201/9780367423926-10)
- Suganya, S., **Sakib, M.**, & Malini, M. *Mitigating adversarial attacks in industrial cybersecurity through NLP strategies for e-commerce: challenges and emerging trends*. In *AI Advancements in Internet of Things and Intelligent Devices*. IGI Global (2025). [doi:10.4018/979-8-3373-7503-8.ch003](https://doi.org/10.4018/979-8-3373-7503-8.ch003)
- **Sakib, M.**, Khan, M. *A Hybrid Soft Computing Framework for Intelligent Time-Series Forecasting Using STL Decomposition and Bayesian-Optimized Deep Learning*. Accepted for publication as a book chapter in *Soft Computing for Web Intelligence: Theory and Applications*, CRC Press (Taylor & Francis Group).
- Suganya, S., **Sakib, M.**, & Khan, M. *Next-Gen Cybersecurity: Integrating AI Devices and NLP for Industrial E-Commerce Defense*. Submitted book chapter.

Edited Books

- *AI Advancements in Internet of Things, Smart Healthcare, and Intelligent Devices*. Edited by Muneer Khan, Shantanu Awasthi, **Mohd Sakib**, and Mohd Vaseem Khan. IGI Global (2025). ISBN13: 9798337357270 | EISBN13: 9798337357294. [doi:10.4018/979-8-3373-5727-0](https://doi.org/10.4018/979-8-3373-5727-0)
- *AI-Driven Security and Intelligence in Cloud and Internet of Things Systems*. Edited by **Mohd Sakib**, Muneer Khan, Shantanu Awasthi, and Mohd Vaseem Khan. IGI Global (2025). ISBN13: 9798337375038 | EISBN13: 9798337375052. [doi:10.4018/979-8-3373-7503-8](https://doi.org/10.4018/979-8-3373-7503-8)
- *Next-Generation AI for Multi Domain Industry Applications*. Edited by Mr. Muneer Khan, Dr. **Mohd Sakib**, Prof. Shishir Kumar, and Prof. Ashutosh Kumar Bhatt. CRC Press (Taylor & Francis Group). **Status:** Proposal Accepted.

Patent / Design Application

- **AI Enabled Wearable Device for Blind Individuals**. Design application record dated 07 July 2026. Applicants/owners: Muneer Khan, **Mohd Sakib**, and Vinay Rishiwal.

TEACHING EXPERIENCE

- Data Structures and Algorithms Theory/Lab – MCA 1st Sem 2024
- Deep Learning – MCA 3rd Sem 2024
- Artificial Intelligence Theory/Lab – MCA 2nd Sem 2025
- Design and Analysis of Algorithms B.Sc 3rd Sem 2025
- Linux Laboratory B.Tech 1st Sem 2025
- Data Structures and Algorithms B.Tech 2nd Sem 2026
- Statistics and Analysis B.Tech 6th Sem 2026

BRIEF RESEARCH SUMMARY

Thesis Project Title

An Efficient Ensemble Deep Learning Model for Estimation and Forecasting of Time Series Data

This project was part of my thesis, and I led most of the work from start to finish. I was actively involved in developing the research idea, designing and implementing the model, conducting experiments, analyzing the results, and preparing the manuscript.

- Proposed a decomposition-driven hybrid forecasting framework integrating CEEMDAN with deep temporal convolution and sequence learning architectures to model complex, nonlinear, and non-stationary time series dynamics.
- Developed a VAR-LSTM fusion model that combines statistical vector autoregression with deep sequence modeling to capture both linear and nonlinear interdependencies in multivariate time series data.
- Designed a genetic algorithm-optimized ensemble model for dynamic feature selection and forecasting performance enhancement under volatile temporal conditions.
- Explored the integration of attention mechanisms, autoencoder-based sequence modeling, and hierarchical deep learning structures to capture long-term temporal dependencies and latent representations in sequential data.

Current Project

E³-TSF: A Dynamic Gating Ensemble Transformer Framework for Multi-Domain Time Series Forecasting

- Designing and implementing a context-aware ensemble transformer architecture that unifies temporal patch learning and variable-wise interdependence modeling within a dynamic gating framework. The proposed model employs a learnable gating network to adaptively allocate weights between complementary transformer backbones based on local time-series characteristics such as trend behavior, volatility, and cross-variable interactions.
- Designing and implementing a context-aware ensemble transformer architecture that unifies temporal patch learning...
- The framework is being evaluated across heterogeneous domains including energy (ETTh1), traffic (METR-LA), finance (Exchange Rate), and epidemiological forecasting (ILI), demonstrating consistent improvements over standalone transformer models and static ensemble approaches.

ACHIEVEMENTS & AWARDS

- Qualified the UGC NET Examination in Computer Science twice (December 2018 and December 2019)
- Awarded Life Membership of the Indian Society for Technical Education (ISTE) (Since 2025)
- Member, IEEE and IEEE Computer Society.
- Awarded the UGC Non-NET Fellowship by the University Grants Commission (UGC), India, for doctoral research and received the fellowship throughout the Ph.D. program
- Recipient of the University Merit Scholarship (2017–2018)
- Received an Appreciation Letter from the Head of the Department for Excellence in Teaching

ACADEMIC ACTIVITIES

- Worked as **Criterion In-Charge** for National Board of Accreditation (NBA) – Documentation and Compliance Activities.
- Served as **Coordinator** for the AICTE PARAKH Initiative and Students' Holistic Development Activities.
- **Co-Convenor**, Five-Day Online Faculty Development Programme (FDP) on “Empowering Minds through AI: Using Innovative Strategies for Effective Problem Solving”
- **Organizer**, Five-Day National Level Online Faculty Development Programme (FDP) on “Innovative Applications of Artificial Intelligence in Cyber-Security,”

- **Member of Editorial Board & Technical Reviewer**, Book titled *Advancing Cybersecurity in Smart Factories Through Autonomous Robotic Defenses*, IGI Global Scientific Publishing, February 2025. ISBN: 979-8-3373-0583-7 | DOI: 10.4018/979-8-3373-0583-7.
- Actively participated in several short-term courses, workshops, and Faculty Development Programmes (FDPs) to strengthen knowledge and practical skills in time series forecasting, deep learning models, transformer architectures, and optimization techniques.

REVIEWING SERVICES

- Computers in Biology and Medicine, Elsevier.
- Cluster Computing, Springer.
- Signal, Image and Video Processing, Springer.
- Electrical Engineering, Springer.
- Scientific Reports, Nature Portfolio.
- International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems, World Scientific.
- Technical Program Committee (TPC) Member at ICIESTR-2024, IIRA 5.0, ICECI 2026.

REFERENCES

Prof. Suhel Mustajab

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Prof. Tamanna Siddiqui

Professor, Department of Information Technology
College of Computer and Information Sciences
Imam Mohammad Ibn Saud Islamic University (IMSIU), Riyadh, KSA
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Dr. Shafiqul Abidin

Associate Professor, Department of Computer Science
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Declaration

I hereby declare that the above-mentioned information is correct up to my knowledge, and I bear the responsibility for the correctness of the above-mentioned particulars.

Place: Dehradun

Date: July 16, 2026

(Mohd Sakib)